

Date	29 September 2026	Team ID	SB_GENAI_042
Project Name	PocketSmart AI	Maximum Marks	5 Marks

PocketSmart AI Technical Documentation & Testing Approach

PocketSmart AI is a GenAI-powered, cross-platform budget & recommendation assistant built using FastAPI/Flask backend and Google's Gemini 1.5 Flash Pro. It provides personalized, budget-aware recommendations across Home Interior, Party Planning, and Jewelry selection domains.

Step 1: System & Testing Overview

Field	Details
Testing Tool / Framework Used	FastAPI / Flask, Pytest, Locust, Postman API Testing
Type of Testing	Functional Testing, API Load Testing, Multimodal Integration Testing
Target Modules / Endpoints	/generate-home, /generate-party, /generate-jewelry, /login, /register
Test Environment	Local Development Server / Google Cloud Platform (Gemini 1.5 Flash Pro API)
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Scenario 1: Home Interior Planning Submit total budget, room types (Living Room, Kitchen), and item quantities (lights, fans).	20	60	Valid JSON with itemized cost breakdown under budget and dynamic shopping links.
2	Scenario 2: AI Party Budget Allocation Submit event type (Wedding/Birthday), guest count, venue type, and party needs.	30	60	Proportional allocation across catering, decor, and entertainment in INR.
3	Scenario 3: Multimodal Jewelry Planner Submit budget, occasion, preferences, and upload an outfit image.	15	45	Image analysis with color coordination and tailored jewelry suggestions.
4	Scenario 4: User Auth & Session Control Execute concurrent login, user registration, JWT token generation, and logout.	50	30	Secure access token issuance and

				seamless session management.
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Step 3: Test Results & Metrics

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2.5 seconds	1.85 seconds	Pass	Gemini API optimized
2	Response Time (Max)	< 5.0 seconds	3.40 seconds	Pass	Multimodal payload
3	Error Rate	< 1%	0.2%	Pass	Edge-cases handled
4	CPU Utilization	< 80%	42%	Pass	Asynchronous endpoints
5	Memory Utilization	< 80%	58%	Pass	Efficient session storage

Step 4: Observations & Analysis

Key Findings:

The PocketSmart AI system successfully generated tailored, budget-compliant recommendations across all three planner domains (Home, Party, Jewelry). Gemini 1.5 Flash Pro demonstrated exceptional accuracy in strict JSON output parsing, budget allocation rules, and multimodal outfit color coordination.

Bottlenecks Identified:

High latency was initially observed during multimodal request handling for the Jewelry Planner when uploading uncompressed high-resolution outfit images. Additionally, external shopping link generation required strict URL encoding validation to avoid broken redirect queries.

Optimization Steps Taken:

- 1. Implemented client-side and server-side image compression using PIL (Pillow) before calling Gemini multimodal API.
- 2. Refined system prompts with JSON schema templates to guarantee structured API responses.
- 3. Added fallback recommendation mechanisms when third-party scraping/search terms return sparse data.

Step 5: Screenshots / Evidence

[Screenshot 1: PocketSmart AI - Architecture & Backend API Pipeline Execution]

[Screenshot 2: PocketSmart AI - User Dashboard & Multimodal Jewelry Recommendations Result]